

Team Name : 64 bits

Ministry /Organization Name / Student Innovation

- Ministry of Education's Innovation Cell (MIC)

PS Code:

- 1592

Problem Statement Title:

- Student Innovation

Team Leader Name:

- Jai Bansal

Institute Name:

- Netaji Subhas University Of Technology

Theme Name:

- Clean & Green Technology



PROBLEM :

Inadequate waste management and increasing potholes, especially during monsoons, lead to health risks, accidents, and road blockages.

Inconvenient Complaint Process: Filing complaints with MCD is cumbersome, requiring physical visits to offices, discouraging many from reporting issues.

Unresolved Issues & Uncertainty: Even when complaints are filed, delays and uncertainty in resolution leave issues unresolved, worsening living conditions.



NEED FOR SOLUTION

Health and Environmental Hazards: Unmanaged garbage attracts disease-spreading pests and contaminates soil and water, harming ecosystems and public health.

Social and Economic Impact: Accumulated waste degrades community life, deters tourism, lowers property values, and diverts municipal resources from essential services like infrastructure and healthcare.

Road Safety Concerns :Accumulated waste degrades community life, deters tourism, lowers property values, and diverts municipal resources from essential services like infrastructure and healthcare.

PROPOSED SOLUTION:

Introducing **SUDHAAR**, our dedicated platform committed to fostering a sustainable and eco-friendly future. This is our contribution to society, empowering individuals and communities to make impactful changes for the betterment of our environment.

How does it help?

- **User/MCD Login:** Individuals and MCD officials can sign in to the SUDHAAR platform.
- **Report Issues:** Users can photograph areas with garbage or potholes that need attention.
- **Upload:** The images are uploaded to the SUDHAAR app.
- **Image Verification:** OpenCV image processing confirms the validity of the issue.
- **Complaint Submission:** A complaint is automatically sent to the MCD for action.
- **Notification:** Users are notified once the issue is resolved.
- **Repost Option:** If the problem remains unresolved for 5 days, a repost option becomes available to escalate the issue.



SUDHAAR

Video link to the demo Prototype (AI Model) : <https://youtu.be/Nm7R3wxflZk>

Key areas of action

Promotions

SUDHAAR actively encourages the use of sustainable products by offering exclusive vouchers for eco-friendly goods and services.

Strategic Ads

The platform features strategically placed advertisements that highlight green initiatives and environmentally responsible businesses, helping users make informed, eco-conscious choices..

ChatBot

An intelligent chatbot designed to address user FAQs, providing quick and accurate responses to common questions about the SUDHAAR platform, from issue reporting to eco-friendly initiatives.

Community

A community-driven discussion forum featuring image sharing, likes, and reactions, enabling users to engage, collaborate, and share insights on sustainability and local issues.

Aadhar Verification

We offer Aadhaar verification to ensure secure and authentic user identification on our platform.

Vouchers & Points

Users earn points for every valid photo and complaint submitted. Accumulated points can be redeemed for vouchers or donated to an NGO, incentivizing active participation and community support.

BUSINESS MODEL

Employment

NGOs can register on the platform to access job opportunities and increase their visibility, expanding their reach and impact within the community.

Advertisement Revenue

Generate income through strategically placed ads promoting eco-friendly initiatives and responsible businesses.

Donation Integration

Allow users to donate accumulated points to NGOs, fostering community support and enhancing social impact..

User Engagement

Users earn rewards for reporting valid issues, encouraging active participation and frequent use of the platform.

Directly address issues by seamlessly connecting users with relevant authorities for swift and effective problem resolution

Featuring a dedicated job portal and volunteer portal for NGOs, allowing organizations to post job openings and recruit volunteers, while also enabling users to discover and engage with meaningful opportunities to contribute.

Effective communication is ensured by making it easy for users to lodge complaints, while authorities receive prompt notifications. Users are then informed as soon as the issue is resolved, streamlining the entire process.

Ultimately, users can effortlessly see their problems resolved, contributing to cleaner neighborhoods and smoother roads, leading to a more effective and sustainable community.

USP



TECH - STACK

React.js

We use React.js to build a dynamic and responsive user interface. It allows us to create reusable components and ensures efficient updates through its virtual DOM.

firebase

We rely on Firebase for backend services, including authentication and real-time database. It provides real-time data synchronization, secure user authentication, and easy integration with other services.

socket.io

We utilize Socket.io for real-time, bidirectional communication between clients and servers. It supports real-time event-based communication, multiple transport protocols, and automatic reconnection.

tenserflow.js

We leverage TensorFlow.js to perform machine learning and AI tasks directly in the browser. It enables in-browser model training and inference, and we can use pre-trained models for tasks like image recognition.

Express.js

We employ Express.js to develop server-side applications and APIs efficiently. It simplifies routing, supports middleware, and integrates seamlessly with various databases and authentication mechanisms.

Node.js

We use Node.js as our runtime environment for executing JavaScript on the server side. Its non-blocking, event-driven architecture helps us handle multiple connections effectively, and the npm ecosystem provides a wide range of development packages.

SUMMARY:

Sustainable Future Focus: **SUDHAAR** is a platform dedicated to promoting a sustainable and eco-friendly environment by addressing local issues like waste management and road maintenance.

Efficient Complaint Resolution: **SUDHAAR** streamlines the complaint process, making it more accessible and transparent, ensuring quicker and more effective solutions to public issues.

Collaborative Efforts: **SUDHAAR** fosters collaboration between citizens and MCD officials, creating a shared responsibility for improving community living conditions and promoting sustainability.



THANKS



